Mehmet Ergezer
Wentworth Institute of Technology
550 Huntington Avenue
Boston, MA 02115, USA
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Editors
Mathematical Programming Computation

Dear Editors,

Please consider the manuscript *"Salem–Spencer sets as a cross-solver benchmark: decomposition, certification, and split-policy effects at $r_3(212)$"* for publication in *Mathematical Programming Computation*.

The manuscript presents a reproducible computational study and benchmark family built from the open frontier decision problem for OEIS A003002. It does not claim to resolve $r_3(212)$. Instead, it develops and evaluates a witness-informed cube decomposition, quantifies the effect of problem-specific window-cardinality cuts, compares CP-SAT, LP/MIP, and CDCL formulations, and releases a graded instance library with machine-checkable proof artifacts.

The study contains five results aligned with MPC's emphasis on practical computation, comparative testing, reusable software, and verification:

- Six monolithic 24-hour baselines fail. A matched five-policy ablation exposes a cover-size versus conquer-cost tradeoff: global AP-degree reduces the exact emitted cover from 12,582,912 to 96,847 cubes, but leaves individually harder subproblems under the historical CP-SAT cap. A standalone C implementation independently reproduces the complete cover.
- A two-stage CDCL survey closes 6,045 of 6,071 broad residuals (99.57%) with no feasible result, while 26 released instances remain as a sharply defined challenge tier.
- On 20 byte-identical hard-pocket formulas run sequentially on matched AMD EPYC 7763 nodes, CaDiCaL 3.0.0 and kissat 4.0.4 both close every instance. Kissat wins all 20 pairs, with geometric-mean CaDiCaL-to-kissat time ratio 1.477 (bootstrap 95% interval 1.332–1.655).
- Six solve-time-stratified survivors from the alternative global-degree cover reproduce the survey CNF hashes, verify through DRAT-to-LRAT conversion, and are accepted by the formally verified `cake_lpr` checker. Separate end-to-end regressions recover the known exact values $r_3(80) = 22$, $r_3(90) = 24$, and $r_3(100) = 27$ from independently verified lower witnesses and `cake_lpr`-checked upper certificates.
- Optimization-form HiGHS experiments show that window inequalities reduce the T1b node count by approximately 81% and tighten every tested LP and MIP upper bound to the decision threshold 44, but not below it.

Source code and benchmark generators are available at `https://github.com/memoatwit/erdos_1194/tree/mpc-submission-v1.2/erdos_r3`. Versioned artifacts are archived at `https://doi.org/10.5281/zenodo.20463334`. The journal-submission version of the archive includes formulas, solver outputs, proof objects, hashes, an independent model audit, and machine-readable provenance, including the six global-degree certificate bundles. The preprint is available as arXiv:2606.04016. The manuscript is not under consideration at another journal.

For Technical Editor review, `ARTIFACT_REPRODUCIBILITY.md` maps each claim to its data and reproduction command. The $N = 80$ exact-value regression is the fastest end-to-end route and runs in minutes on a workstation; the archived certificates can also be checked with `drat-trim` and `cake_lpr` without re-solving the instances.

The paper retains and explicitly refutes a working hypothesis formed during the campaign. We believe this is methodologically useful: it demonstrates why unresolved instances, solver configurations, and proof objects should be archived and retested under controlled pipelines.

Generative-AI use is disclosed in the manuscript. The author made all final scientific and experimental decisions, independently checked the numerical and mathematical claims, and takes full responsibility for the work.

Sincerely,

Mehmet Ergezer
Wentworth Institute of Technology
ergezerm@wit.edu
ORCID: 0000-0001-6627-3667